Pandelis Zembashis

CTO | Autonomous Workflow Engineer | LLM Orchestration, Retrieval & Agent Systems

p@ndel.is | pa.ndel.is | github.com/PandelisZ | x.com/PandelisZ | linkedin.com/in/Pandelis

EXECUTIVE SUMMARY

CTO and hands-on engineer building autonomous LLM workflows, custom agent frameworks, retrieval/memory systems, and distributed execution platforms. At Cosine, lead product and engineering for an applied AI lab where users drive complex software-engineering tasks through natural-language conversation: browser research, repository retrieval, memory, tool use, long-running plans, subagents, deterministic replay, evaluation traces, and secure execution. Strong fit for Latent Labs' scientific agent role: translating frontier model capability into reliable reasoning workflows, APIs, knowledge discovery, and intuitive chat interfaces for expert users.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Applied AI lab building autonomous coding agents, natural-language workflows, retrieval, and secure runtimes.

- Lead product and engineering for Cosine's AI coding platform, turning natural-language requests into long-running autonomous workflows across codebases, browsers, terminals, files, and external tools.
- Built Cosine's custom agent harness rather than wrapping another framework: planning, streaming tool calls, checkpoints, deterministic replay, memory writes, MCP/LSP integration, subagents, file locks, and structured rollout logs.
- Built retrieval and context systems for large codebases: repository ingestion, native search, memory, structured file reads, browser/internet research, and scoped context assembly for multi-step reasoning.
- Designed workflow orchestration for multi-agent software engineering, including background subagents, task state propagation, evaluation traces, human review, and parallel sandboxed runs.
- Built secure execution from 2024 onward: Kubernetes-scheduled remote agents, gVisor/runsc isolation, ephemeral per-session environments, network-policy boundaries, and safe shell/tool access.
- Shipped one runtime across CLI, desktop, VS Code/WebAssembly, API-server sessions, and remote worker pods, keeping chat-driven workflows consistent across product surfaces.
- Worked with users and internal research/product needs to convert ambiguous workflow pain points into product primitives, evaluation signals, and public technical demos.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled event-driven automation systems that processed millions of email chains and billions of tokens across natural-language sales workflows.
- Established engineering cadence, incident response habits, and onboarding rhythms during rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the Ruby on Rails backend by introducing GraphQL APIs, improving product iteration across web, operations, and marketplace workflows.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, incident response, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021
White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

WRITING & PUBLIC WORK

Why we built our own harness: Technical deep dive on Cosine's custom agent harness, tool execution, replay, multi-agent orchestration, gVisor/Kubernetes runtime, and ML feedback loop.
Cosine Swarm: Long-horizon agent orchestration, parallel workflows, and coordination patterns for autonomous software engineering.
Parallelising Development: Public writing on breaking complex goals into concurrent agent workflows and productizing multi-agent output.

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

LLM Orchestration & Agents: Custom agent frameworks, tool use, planning, subagents, memory systems, deterministic replay, OpenAI/Anthropic APIs, MCP/LSP
Retrieval & Knowledge Workflows: Repository ingestion, semantic/context assembly, browser research, native search, structured documents, citations, long-context workflows
Workflow & Distributed Systems: Go, Python, TypeScript, APIs, event-driven architecture, Kubernetes, gVisor/runsc, WebSockets, queues, observability
Product Engineering: Chat interfaces, developer experience, research-to-product loops, technical writing, enterprise deployment, secure execution, evaluation traces